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Project Description

NCBI hosts and manages the Database of Genotypes and Phenotypes (dbGaP) and NIH's Sequence Read Archive (SRA). dbGaP provides and manages access to protected data related to human studies that have investigated the interaction of genotype and phenotype. In partnership with the NIH Science and Technology Research Infrastructure for Discovery, Experimentation, and Sustainability (STRIDES) Initiative, NCBI has made the entire corpus of SRA and computational tools accessible on the cloud (commercial and open access) in addition to NCBI's local servers.

NCBI is actively developing technical solutions that modernize and streamline secure access to controlled-access data via GA4GH Data Repository Service (DRS) and NIH Researcher Authorization Service (RAS).

The central goal is to create an equitable and interoperable ecosystem where NIH-funded data is FAIR (findable, accessible, interoperable, and reusable) and NCBI is also an engaged partner in the development of community-driven solutions to provide secure access to protected data in the federated data access landscape.

Participants

NCBI
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ODSS coordination support

NIH Institutes and Centers
NIH GDS Taskforce including representatives of IC Genomic Program Administrators and Data Access Committees

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dbGaP Data

Studies	1,865
Subjects	~2.9 Million
Samples	~3.4 Million
Phenotype: Variables	370,825
Values	~2.5 Billion
Study Documents	7,120
Association Analyses	7,883
Genotype Assays (array)	~2 Million
Genotype Assays (imputed)	543,137
Genotype Assays (seq derived)	399,269
Sequence (WGS SRA)	178,288
Sequence (WXS SRA)	271,447
Sequence (RNAseq SRA)	86,879
Epigenomic (SRA)	~35,000

See: dbGaP Summary Stats; numbers change daily

SRA Data

Public Sequence Data (Number of Records)						
Data Format	**Google Cloud Platform**			**Amazon Web Services**		
	Public Dataset (Hot)	Commercial (Hot)	Commercial (Cold)	Open Data Program (Hot)	Commercial (Hot)	Commercial (Cold)
Source	0	0	14.2M	0	0	14.2M
SRA Normalized	775,619	7.5M	5.9M	13.4M	825,126	0
SRA Lite	0	8.0M	0	0	0	0
Controlled-Access Sequence Data (Number of Records)						
Source	0	0	749,915	0	0	749,801
SRA Normalized	0	608,671	141,244	0	705,101	44,700
SRA Lite	0	1,751	0	0	0	0

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Services

Search
- Entrez/ SOLR indexing public metadata
- Cross linking accessions across dbGaP, BioProject, BioSample and SRA dbs

Request
- dbGaP Controlled Access system:
- PI Selection/ Reporting
- DAC Review

Data Access/Download
- RAS Clearinghouse (CLR): Performs auditing and validation of RAS passport tokens and dbGaP permissions on behalf of RAS clients.
- IDX: Performs object id exchange between SRA INSDC-style accessions and DRS ids.
- DRS: Resolves DRS ids to object location and validates access authorization via Clearinghouse.
- FHIR API: Provides access to dbGaP study level metadata

Tools

SRA Data Locator (SDL): supports finding data in appropriate locations

SRA Toolkit: SRA Toolkit supports retrieval and conversion of data from into requested file format (FASTQ, ...)

Cloud Data Delivery Service (CDDS): Service to request data in cold storage to be delivered to researchers' cloud bucket.

ElasticBLAST: Handles large sequence-based queries. Cloud native, Alpha versions on AWS and GCP.

BLAST+ in Docker: Cloud-based BLAST.

A Comprehensive list of NCBI Tools can be found here: All Resources - Site Guide - NCBI (nih.gov)

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Authentication

Prospective users must have NIH eRA, NIH Login or GSA login.gov account.

RAS IDs are used as an authentication mechanism for controlled-access data and to gain access to dbGaP authorizations.

Authorization

Authenticated users submit Data Access Requests through dbGaP Controlled Access system.

Approvals are delivered to RAS as pre-authorizations to access data. These are encapsulated within RAS passport tokens.

DRS approves requests for data access based upon consultation with the RAS Clearinghouse and permissions within RAS passport token.

Indexing

Data objects are assigned permanent globally unique NCBI Accessions to allow for Access or Download across tools

Data cross-links are maintained across dbGaP, BioProject, BioSample and SRA

Datasets are identified through faceted search for public object-level metadata

Data Models

The data in dbGaP are organized as a hierarchical structure of studies. Accessioned objects within dbGaP include studies, phenotypes (as variables and datasets), various molecular assay data SNP and Expression Array, Sequence, and Epigenomic marks, analyses, and documents

See: dbGaP Data Model

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Architectural Diagram – planned end state